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April 5, 1967

SUPERSEDING
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FEDERAL SPECIFICATION

PLATING, CADMIUM (ELECTRODEPOSITED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for electrodeposited cadmium plating.

1.2 Classification.

1.2.1 *Classes.* Cadmium plating covered by this specification shall be of the following classes, as specified (see 6.2):

1—0.00050 inch thick.
2— .00030 inch thick.
3— .00020 inch thick.

1.2.2 *Types.* The cadmium plating shall be of the following types, as specified (see 6.2):

I—Without supplementary chromate or phosphate treatment.
II—With supplementary chromate treatment (see 3.3.3 and 6.1.2).
III—With supplementary phosphate treatment (see 3.3.4 and 6.1.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The materials used shall be such as to produce coatings which meet the requirements of this specification.

3.2 Workmanship.

3.2.1 *Basis metal.* The basis metal shall be free from defects that will be detrimental to the appearance or the protective value of the plating. The basis metal shall be subjected to such cleaning, pickling, and plating procedures as are necessary to yield deposits as hereinafter specified.

3.2.2 *Plating.* The cadmium deposit shall

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be smooth, fine-grained, adherent, and free from blisters, pits, nodules, indications of burning, and other defects. Superficial staining which has been demonstrated as resulting from rinsing, or slight discoloration resulting from baking as specified (see 3.3.6), shall not be cause for rejection. All details of workmanship shall conform to the best practice for high quality plating.

3.3 General requirements.

3.3.1 Unless otherwise specified (see 6.2), the plating shall be applied after all machining, brazing, welding, forming, and perforating of the article have been completed.

3.3.2 Unless otherwise specified (see 6.2), steel parts having an ultimate tensile strength of 220,000 p.s.i. or above shall not be plated without specific approval of the procuring activity.

3.3.3 Unless otherwise specified (see 6.2), the chromate treatment required for conversion to type II shall be either a chemical or electrochemical treatment in an aqueous solution of salts, acids, or both, and containing hexavalent chromium and activating acid radical(s) to produce a continuous smooth, distinct protective film, distinctly colored iridescent bronze to brown including olive drab and yellow. The articles so treated shall be thoroughly rinsed and dried according to requirements for the particular chemical or electrochemical process used. Type II plating shall be similar in appearance to coatings on separate specimens which are capable of passing the salt-spray test (see 3.4.2, 4.4.2.3, and 4.5.3). Usual chromic and nitric acid bright dips for cadmium are not chromate treatments.

3.3.4 Unless otherwise specified (see 6.2), the phosphate treatment required for conversion to type III shall be a chemical treatment in an acqueous solution of phosphoric acid, phosphates and accelerating chemicals to produce a tightly adherent film (see 3.4.3).

3.3.5 All ferrous parts shall be given a stress relief heat treatment at 375°F. for 3 hours prior to cleaning and plating if they contain objectionable residual stresses (see 6.9).

3.3.6 All steel parts having a hardness of Rockwell C40 and higher shall be baked at 375° ± 25°F. (191° ± 14°C.) for three hours or more after plating. Plated springs or other parts subject to flexure shall not be flexed prior to the baking operations. In the case of types II and III treated parts, the baking treatment shall be done prior to the application of the supplementary coatings. Cadmium-plated surfaces passivated as a result of the baking operation shall be reactivated prior to receiving the type II supplementary chromate treatment (see 6.6).

3.3.7 Cadmium shall be deposited directly on the basis metal without a preliminary plating of other metal, such as nickel, except in the case of parts made of corrosion-resisting steel on which a preliminary plating of nickel is permissible.

3.3.8 When aluminum is the basis metal the cadmium plating process shall be as specified by the procuring agency.

3.3.9 Unless otherwise specified (see 6.2), either a bright or dull finish will be acceptable.

3.4 Detail requirements.

3.4.1 *Thickness of plating.* Unless otherwise specified (see 6.2), the thickness of cadmium shall be as specified in table I on all visible surfaces which can be touched by a ball 0.75 inch in diameter. Where class 1 is specified, all other visible surfaces shall be class 2 minimum thickness. Where class 2 is specified, all other visible surfaces shall be class 3 minimum thickness. Where class 3 is specified, all other visible surfaces shall be not less than 0.00015 inch minimum thickness.

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TABLE I. *Thickness*

Class	Thickness, minimum, inch
1	0.00050
2	.00030
3	.00020

3.4.1.1 Unless otherwise specified, cadmium plating supplied for weapons systems shall be class 1 thickness and for use on surfaces to be painted shall be type III, unless otherwise specified or controlled by the following exceptions (see 6.1.3, 6.2, and 6.7).

(a) Bolts, studs, washers, and articles with portions externally threaded. These parts shall have a minimum of class 2 thickness.

(b) Holes and other openings, and internal threads from which the external environment is completely excluded shall not be subjected to a thickness requirement.

3.4.2 *Salt spray requirements.* Type II treatment shall show no white corrosion products of cadmium as defined in 3.4.2.1 at the end of 96 hours, when tested by continuous exposure to the salt spray in accordance with 4.5.3.

3.4.2.1 For type II treatment, the appearance of white corrosion products, visible to the unaided eye at normal reading distance, at accidental scratches through the chromate film to the plate, or at unscratched areas of the chromate film shall be cause for rejection, except that white corrosion products at edges of specimens shall not constitute failure.

3.4.3 *Water resistance.* The supplementary phosphate treatment (type III) shall afford a good base for paint. Paint coatings shall show no blistering following subjection to the water-resistance test described in 4.5.4.

3.4.4 *Adhesion.* The adhesion of the plating shall be such that when examined at a magnification of approximately 4 diameters, it does not show separation from the basis metal at the interface when subjected to the tests described in 4.5.2. The interface between the cadmium and the basis metal is the surface of the basis metal before plating. The formation of cracks in the plate caused by rupture of the basis metal which do not result in flaking, peeling, or blistering of the plate shall not be considered as nonconformance to this requirement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Separate specimens. When the plated articles are of such form as to be not readily adaptable to a test specified herein, or are for destructive tests of small lot sizes, the test shall be made by the use of separate specimens plated concurrently with the articles represented. The separate specimens shall be of a basis metal equivalent to that of the articles represented. "Equivalent" basis metal includes chemical composition and finish of surface prior to plating. For example, a cold-rolled steel surface should not be used to represent a hot-rolled steel surface; due to the impracticality of forging or casting separate test specimens, hot rolled steel specimens may be used to represent forged and cast-steel articles. The separate specimens shall be strips approximately 1 inch wide, 4 inches long, and 0.04 inch thick, for adhesion tests, but shall be at least 4 inches wide, 6 inches long, and approximately 0.04 inch thick for all other tests. These specimens shall be introduced into a lot at regular intervals prior to the cleaning operations preliminary to plating

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and shall not be separated therefrom until after completion of the processing. Conditions affecting the plating of the specimens including the spacing and positioning with respect to anodes and to other objects being plated shall correspond as nearly as possible to those affecting the significant surfaces of the articles represented. Separate specimens shall not be used for thickness measurements, however, unless the necessity for their use has been demonstrated.

4.3 Lot. A lot shall consist of plated articles of the same type and class plated under the same conditions, and approximately the same size and shape submitted for acceptance at one time.

4.4 Sampling.

4.4.1 *For visual examination and nondestructive tests.* Plated articles shall be sampled in accordance with level II in MIL-STD-105. The acceptable quality level (AQL) shall be 4.0 percent defective in accordance with MIL-STD-105. The lot shall be accepted or rejected according to the procedures in 4.4.1.1 for visual examination and 4.4.1.2 for plating thickness (nondestructive tests).

4.4.1.1 *Visual examination.* Samples selected in accordance with 4.4.1 shall be examined for compliance with the requirements of 3.2.1 before plating and 3.2.2 after plating. If the number of nonconforming articles exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.1.2 *Thickness of plating (nondestructive tests).* Samples selected in accordance with 4.4.1 shall be inspected and the plating thickness measured at locations on each article as defined in 3.4.1 for compliance with the requirements. The article shall be considered nonconforming if one or more measurements fail to meet the specified minimum thickness. If the number of defective items in any sample exceeds the acceptance number for the specified sample, the lot represented by the sample shall be rejected.

4.4.2 *For destructive tests.* A random sample of four plated articles shall be taken from each lot or four separately plated specimens shall be prepared in accordance with 4.2 to represent each lot. If the number of the articles in the lot is three or less, the number of the articles in the sample shall be specified by the procuring agency (see 6.2).

4.4.2.1 *Thickness of plating (destructive tests).* Samples selected in accordance with 4.4.2 shall be measured for plating thickness at locations on each article as defined in 3.4.1 for compliance with the specified requirements. If the plating thickness at any place on any article or specimen is less than the specified minimum thickness, the lot shall be rejected.

4.4.2.2 *Adhesion.* The articles or specimens used for the destructive thickness test (see 4.4.2.1), if of suitable size and form, may be used as the specimens for the adhesion test to determine compliance with the requirements of 3.4.4. Failure of one or more of the test specimens shall constitute failure of the lot.

4.4.2.3 *Salt spray.* Unless otherwise specified (see 6.2), compliance with the requirements of 3.4.2 shall be determined for type II treated articles. A set of four separate test specimens, prepared in accordance with 4.2 in lieu of the plated articles, shall be tested. Failure of one or more of the test specimens shall reject the lot.

4.4.2.4 *Water resistance.* Unless otherwise specified (see 6.2), compliance with the requirements of 3.4.3 shall be determined for type III treated articles. A set of four separate test specimens, prepared as described in 4.2 in lieu of the treated plated articles, shall be tested. Failure of one or more of the test specimens shall reject the lot.

4.4.2.5 *Embrittlement relief.* Unless otherwise specified (see 6.2), a random sample of four plated articles or parts shall be taken from each lot. Samples shall be tested

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for conformance to the requirements of 3.3.6. The samples shall be considered to be nonconforming if cracks are found after test or fail by fracture. Failure of one or more test articles shall reject the lot.

4.5 Test methods.

4.5.1 *Thickness.* Either method 520, electronic test, or method 522, magnetic test, of Fed. Test Method Std. No. 151 may be used for the nondestructive measuring of the cadmium plating thickness. Either method 521, microscopic test, or method 523, chemical dropping test, of Fed. Test Method Std. No. 151 may be used for the destructive measuring of the cadmium plating thickness. For referee test, the microscopic test method shall be used. Thickness measurements of types II and III cadmium shall be made after the application of types II and III supplementary coatings and subsequent removal. The chromate film may be removed from type II plating by using a very mild abrasive (a paste of levigated alumina rubbed on with the finger). The phosphate coating may be removed from type III plating by scrubbing with a rubber policeman in a 10 percent solution of NaOH (usually takes from 10 to 15 minutes).

4.5.2 *Adhesion.* Adhesion may be determined by scraping the surface of the plated article to expose the basis metal and examining at four diameters magnification for evidence of nonadhesion. Alternatively, the article may be clamped in a vise and the projecting portion bent back and forth until rupture occurs. If the edge of the ruptured plate can be peeled back or if separation between the plate and the basis metal can be seen at the point of rupture when examined at four diameters magnification, adhesion is not satisfactory.

4.5.3 *Salt-spray test.* The salt-spray test shall be made in accordance with method 811 of Fed. Test Method Std. No. 151. To secure uniformity of results, type II supplementary coatings shall be aged at room temperature for 24 hours before subjection to the salt-spray.

4.5.4 *Water-resistance test.* The samples after receiving the supplementary phosphate (type III) treatment with the lot represented shall be coated with 0.7 to 1.2 mils (dry-film thickness) of the primer or enamel intended for use on the article. The coated specimens shall be dried in conformance with the requirements of the coating material, either by air drying for 90 hours or by baking at the required temperature for the required period of time followed by 24 hours air drying. The specimens shall then be immersed in distilled water at pH 5.0 to 7.0 at 75°F., minimum, for 24 hours. The appearance of blisters in the paint coating shall be cause for rejection.

4.5.5 *Embrittlement relief.* Unless otherwise specified (see 6.2), samples of coated parts taken as specified in 4.4.2.5 for determining compliance with 3.3.6 are these which will be subjected to a sustained tensile load in use. Parts such as spring pins, lockrings, etc., which are installed in holes or rods shall be similarly assembled, using the applicable parts specification or drawing tolerances which impose the maximum sustained tensile load on the plated parts. The selected samples shall be subjected to a sustained tensile load equal to 115 percent of maximum tensile load for which the part was designed (design yield load). The articles or parts shall be held under load for at least 200 hours, unless otherwise specified (see 6.2), and then examined for cracks. Parts which require special fixtures or extreme loads to comply with the above requirements shall be tested as specified by the procuring activity.

4.6 **Retests.** Plated articles which have been subjected to the chemical dropping test for thickness (see 4.5.1) and articles rejected or withdrawn because of defective or deficient plating may be resubmitted after stripping and replating or screening of the entire lot. Full particulars concerning

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the replating shall be furnished to the procuring activity. Where only insufficient thickness is involved, replating without stripping the existing coating may be permitted at the discretion of the procuring activity. All material having a hardness greater than Rockwell C40 shall be hydrogen-embrittlement relieved after replating.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification, (see 6.8).

6. NOTES

6.1 Intended use.

6.1.1 The primary purpose of electrodeposited cadmium plating is to provide corrosion protection to ferrous parts.

6.1.2 The primary purpose of chromate finishes is to retard or prevent the formation of white corrosion products on surfaces exposed to stagnant water, high humidity atmospheres, salt water, marine atmospheres, or exposed to cyclic condensation and drying. Some types of chromate coatings have proved satisfactory as a base for paint.

6.1.3 The primary purpose of phosphate finishes is to form a paint base.

6.1.4 Cadmium coatings should not be used on parts which in service reach a temperature of 450°F. or higher, or come in contact with other parts which reach those temperatures.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Class (thickness) required (see 1.2.1 and 3.4.1).
(*c*) Type required (see 1.2.2).
(*d*) Specified application procedures, if applicable (see 3.3.1 through 3.3.4, 3.3.8 and 3.3.9).
(*e*) Test options (see 4.4.2 and 4.4.2.3 through 4.4.2.5).

6.3 Toxicity. Cadmium, because of its toxicity, shall not be employed as a coating for any object intended for use as a food container or cooking utensil or for any object likely to come in contact with food or beverages. Cadmium-coated sheets and any other structural shapes which may be subjected to heat from welding or soldering operations should be so labeled because of the danger from poisonous vapors during such operations.

6.4 Type II. Supplementary chromate treatments may be used to prevent finger marking damage which may occur during assembly and storage.

6.5 Supplementary chromate treatments (type II) which involve only dipping in chemical solutions normally require a sufficient period of drying to render the parts suitable for handling without damage to the coating while in gelatinous form; and it is important with such that the coating is not excessively damaged while wet. Chromating processes of the electrochemical type normally produce conversion films that do not require drying to avoid film fracture in production.

6.6 Surfaces of cadmium plating intended for conversion to type II and which have become passive as a result of the baking operation specified in 3.3.6 may be reactivated by brief immersion in dilute acid. If, for example, the chromating solution is acidified with sulphuric acid, then the reactivating solution should be 1 part sulphuric acid (sp. gr. 1.83) by volume added to 99 parts water, or for further example, if the chromating solution is acidified with hydrochloric acid, then the activating solution should be 1 part hydrochloric acid (sp. gr. 1.16) by volume added to 99 parts water. The duration of immersion should be as brief as is consistent with the nature of the work. For example, a perforated container of barrel-plated parts would be expected to be reactivated in approximately 15 seconds and separately racked items in approximately 5 seconds. The surfaces should be activated

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as soon as possible following baking operation and should be handled carefully to avoid contamination by dirt or grease.

6.7 The dimensional tolerance of most threaded articles, such as nuts, bolts, and similar fasteners with complementary threads do not permit the application of a coating thickness much greater than class 3. If heavier coatings are required for satisfactory corrosion resistance, allowances must be made in the manufacture of the threaded fasteners for tolerance necessary for plate build-up.

6.8 Unprotected cadmium plated articles should not be packed in nonventilated containers, either together or in contact with electrical equipment, because of the danger of deleterious effects on cadmium plating from unstable organic electrical insulation. In addition to organic electrical insulation, phenolic resinous substances and others containing unsaturated carbon to carbon linkages, such as oil paints and impregnated water, etc., cause an abnormal attack on cadmium by setting free in the presence of moisture formic acid, butyric acid, etc. Corrosion of cadmium coatings and the steel basis metal has been noted when cadmium plated articles have been packaged in direct contact with container materials such as wood or cardboard. Corrosion has been especially severe if the container materials have become wet or have been stored under conditions of high humidity.

6.9 There is a hazard that cold-worked or cold-straightened steel parts of hardness Rockwell C 40 and below may crack during cleaning and plating. If experience indicates that this is a problem, such parts should be heat treated at 375° ± 25°F. for 3 hours.

CUSTODIANS:
Army—MR
Navy—WP
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Review activities:
Army—MR, MI, MU, WC, EL
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